Does the Standard Sparse-Autoencoder Feature-Absorption Metric Over-Report for Non-Spelling Features?

A Cross-Metric, Cross-Model Study of Feature Absorption Beyond First-Letter Spelling

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Abstract

Sparse autoencoders (SAEs) are meant to decompose language-model activations into monosemantic, human-interpretable features, but *feature absorption*—in which a seemingly general feature silently fails on a subset of tokens while a token-specific latent quietly carries the concept— undermines that interpretation. Absorption is quantified in the literature almost exclusively on a single family of tasks, "does this token start with the letter X," and with two different metrics: an older *ablation* (behavioral) metric that attributes the model's answer to individual latents, and the current SAEBench-standard *projection* (representational) metric that decomposes a concept direction into per-latent contributions. We ask a question that has, to our knowledge, not been tested: do the two metrics still agree once absorption is measured on a structurally different, non-spelling property? Running both metrics on the binary property is-capitalized across Gemma-2-2b and Gemma-2-9b (Gemma Scope 16k SAEs), we find that they *disagree*. The projection metric reports single-dominant-latent absorption modestly above its random-direction null (0.14, bootstrap 95% CI $[0.085, 0.200]$ vs. 0.04; n=130, a marginal separation), whereas the ablation metric reports exactly zero on 273/273 candidate tokens, on *both* models. Two controls rule out the mundane explanations: the zero is not caused by the model failing the task (Gemma-2-9b performs is-capitalized at 0.91 accuracy, versus 0.60 for 2b, yet ablation absorption stays 0.0) and not by a dead layer (at the same 9b layer the ablation metric fires at 0.045 for spelling, and at *matched* task accuracy it fires at $\sim$0.026 for spelling while is-capitalized remains 0.0). We conclude that the field-standard projection metric appears to over-report single-latent absorption for a distributed structural feature that has no causal correlate. As a secondary, controlled result, we show—through two independent task-accuracy manipulations—that ablation-based absorption is itself sensitive to task performance, whereas the projection metric is not. We reproduce first-letter absorption and its dependence on SAE sparsity as pipeline validation. Effects are near-floor and the study spans one well-powered structural property, one model family, and one SAE recipe; we therefore present the result as a metric-validity caveat, argue that representational absorption signals require causal validation, and specify exactly what would be needed to elevate it to a robust claim. An appendix documents three hypotheses we built controls to test and then rejected.

Contents

1 Introduction

Mechanistic interpretability seeks to reverse-engineer the internal computations of neural networks into human-understandable parts, both as basic science and as a foundation for auditing and steering models whose capabilities are hard to trust by behavior alone. A central obstacle is *superposition*: models appear to represent far more features than they have neurons, packing them into linear combinations of directions rather than into individual units [7]. Sparse autoencoders (SAEs) are the leading proposed remedy: by learning an overcomplete, sparsely-activating dictionary that reconstructs a model's activations, they aim to recover the underlying features as individual, monosemantic latents [1, 6, 21]. Open SAE suites such as Gemma Scope [17] have made this decomposition available at scale and turned SAE features into a common substrate for interpretability tooling.

The value of that substrate rests entirely on the latents meaning what they appear to mean. *Feature absorption*, identified by Chanin et al. [4], is a sharp counterexample. A latent may look like a clean detector for a general concept—say "the token starts with the letter S"—while in fact it silently deactivates on particular tokens (e.g. *short*), and a more specific, token-aligned latent absorbs the general direction and carries the concept for those tokens. The "general" latent is therefore not a reliable detector: it has arbitrary, token-dependent blind spots. Absorption is corrosive precisely because it is invisible to the usual sanity checks—the latent still has high precision and looks monosemantic—and because it appears to be a systematic consequence of how SAEs trade off sparsity against reconstruction when features co-occur [4, 5].

Because absorption matters for whether we can trust SAE features, how we *measure* it matters too. Yet two facts about current practice motivate this paper. First, absorption is measured almost entirely on one task: whether a single-token word starts with a given letter, chosen because it provides clean, objective ground truth. Benchmarks instantiate absorption on first-letter spelling and nothing else [14], and Chanin et al. [4] explicitly flag "examples of feature absorption unrelated to character identification" as open. Second, two different metrics are in use. The original [4] is *causal*: it uses attribution to ask whether a single latent actually mediates the model's answer. SAEBench [14] standardized a *representational* alternative that decomposes the residual stream's projection onto a concept direction into per-latent contributions, adopting it because the causal/ablation signal "goes to zero in later layers." The representational metric is now the field standard; the ablation metric is effectively deprecated.

These two facts combine into a simple, untested question:

> *Do the two standard absorption metrics still agree once absorption is measured off the spelling task, on a structurally different property?*

We study this for objective, non-spelling token properties—primarily is-capitalized, whether a single-token word begins with an uppercase letter—and find that the metrics *disagree*. The representational metric reports (weak, above-null) single-latent absorption for is-capitalized; the causal metric reports a clean zero. We then spend most of the paper ruling out the boring reasons for that zero, because one of them— "the causal metric is known to die in later layers"—is a documented artifact rather than evidence of no absorption. Using a cross-model control (Gemma-2-9b, which performs the task much better than 2b) and a layer- and accuracy-matched spelling control, we show the zero is genuine: it is neither a task-performance artifact nor a dead-layer artifact.

Contributions.

1. **A cross-metric comparison off the spelling task.** We run *both* standard absorption metrics—the SAEBench projection metric and the Chanin ablation metric—on structural token properties and contrast them directly. We are careful not to over-claim novelty: Chanin et al. [5] already run k-sparse probing on part-of-speech, so this is not the first look at non-character features; it is, to our knowledge, the first *direct contrast of the two absorption metrics* on such features (§3).
2. **A metric disagreement, with confidence intervals.** For is-capitalized, projection reports above-null absorption while ablation reports zero on 273 candidate tokens; we report bootstrap CIs and candidate counts throughout, because the effect is near-floor (§5.2, 5.3).
3. **Controls that isolate the cause.** A cross-model task-ability control and a layer- and accuracy-matched spelling control together rule out both "the model cannot do the task" and "the layer is dead" (§5.4).
4. **A reliability caveat for the ablation metric.** Two independent task-accuracy manipulations show ablation-based absorption tracks task performance while the projection metric does not (§5.5).
5. **Reproducibility and honesty.** We release all code, a model-agnostic re-implementation used for the cross-model controls, an independently review-verified single-class metric port, and an appendix (C) documenting three hypotheses we built controls to test and rejected.

We emphasize up front that the result is small: the effects are near-floor, only one structural property is well-powered, and both models come from one family. We treat it as a metric-validity caveat rather than a discovery, and §8 states exactly what would make it robust.

2 Preliminaries and Notation

Sparse autoencoders. Let $\mathbf{x} \in \mathbb{R}^d$ be a residual-stream activation of a transformer at a fixed layer and token position. An SAE with $m \gg d$ latents encodes and decodes

$$\mathbf{a} = \sigma(W_{\text{enc}}\,\mathbf{x} + \mathbf{b}_{\text{enc}}) \in \mathbb{R}^m_{\geq 0}, \qquad \hat{\mathbf{x}} = W_{\text{dec}}\,\mathbf{a} + \mathbf{b}_{\text{dec}}, \tag{1}$$

trained to minimize reconstruction error under a sparsity penalty, so that only a small number $L_0 = \|\mathbf{a}\|_0$ of latents are active per input. We write $W_{\text{dec}}[i] \in \mathbb{R}^d$ for the decoder direction of latent i and a_i for its activation. We use Gemma Scope JumpReLU SAEs [17] on Gemma-2 [10]; the SAEs are folded so that decoder directions are unit norm, with the norm absorbed into a_i, so $\hat{\mathbf{x}} = \sum_i a_i W_{\text{dec}}[i]$ up to bias and reconstruction error.

Concepts, probes, and main latents. A *concept* is a binary or categorical property of the input token with objective ground truth (e.g. "the first letter is S," or "the token is capitalized"). For each concept c we train a linear probe on the residual stream and take its (unit) weight vector $\hat{\mathbf{p}}_c$ as the ground-truth concept direction, following the linear representation hypothesis that concepts are encoded as directions in activation space [7, 20]. Running k-sparse probing over SAE latents identifies the concept's *main latent(s)* $M_c \subseteq \{1, \ldots, m\}$: the small set that a sparse probe uses to predict c, grown by a feature-splitting criterion (extend k while validation F_1 jumps by at least 0.03).

Feature absorption.

Definition 1 (Feature absorption, informal)**.** On a token t whose concept c holds and whose residual-stream probe fires ($\hat{\mathbf{p}}_c \cdot \mathbf{x}_t > 0$), absorption occurs when the main latents M_c are *silent* ($a_i < \varepsilon$ for all $i \in M_c$) yet the concept is nonetheless present, carried by some other, more specific latent. Such tokens are *candidate false negatives*: the general latent misses them.

The two metrics differ in how they decide whether a candidate is genuinely absorbed.

The ablation (behavioral) metric. Chanin et al. [4] operationalize absorption *causally*. On a candidate token, they use integrated-gradient attribution of the SAE latents to a task metric $g(\cdot)$ (a logit difference for the correct answer, e.g. the target-letter logit minus the mean of the other letters). A token is counted as absorbed if (i) the main latents are silent, (ii) a single latent has the most-negative attribution (its ablation most hurts the answer) by a clear margin, and (iii) that latent is aligned with the probe direction. This is a statement about what the model *uses* to produce its answer.

The projection (representational) metric. SAEBench [14] instead decomposes the concept content of the residual. Writing the projection of $\mathbf{x}_t$ onto the concept direction as

$$\hat{\mathbf{p}}_c \cdot \mathbf{x}_t \;=\; \sum_{i=1}^{m} \quad \underbrace{a_i \,(W_{\text{dec}}[i] \cdot \hat{\mathbf{p}}_c)}_{c_i \,:\; \text{latent } i\text{'s contribution}} \quad + \,(\text{bias, error}), \tag{2}$$

a candidate is counted as absorbed if, with the main latents silent, one non-main latent *dominates* the projection. This is a statement about how the concept is *represented*, independent of whether the model behaviorally uses it. SAEBench adopted this form because the ablation effect "goes to zero in later layers," i.e. the causal metric becomes unreliable with depth—a point we return to repeatedly, because it means "ablation reads zero" is, in later layers, a known artifact.

Two knobs we make explicit. Because SAE codes are sparse, at any token the projection (2) is dominated by whichever few latents are active; "one dominant latent" can therefore arise partly from sparsity geometry rather than from concept-specific absorption. We control for this with a *random-direction null* (replace $\hat{\mathbf{p}}_c$ by random unit directions) and with a scale-free *dominance ratio* R: a token is "single-dominant" iff $c_{(1)} \geq R\, c_{(2)}$ for the top two contributions. We report rates at $R \in \{1.5, 2, 3\}$ and condition on the main-latent-silent pool, so the rate does not bake in main-latent quality (Appendix C explains why this matters).

3 Related Work

SAEs and the monosemanticity program. The idea that overcomplete sparse dictionaries can disentangle superposed features [7] was turned into a practical LM tool by Bricken et al. [1] and Cunningham et al. [6], then scaled to frontier models [21] and released openly for Gemma-2 as Gemma Scope [17]. Our work takes these SAEs as given and studies a property of the *evaluation* of their features.

Feature splitting, absorption, and hedging. Chanin et al. [4] define feature splitting (a latent fragments into several as width grows) and the pathological special case of absorption, giving the ablation metric and the first-letter benchmark. Chanin et al. [5] describe *feature hedging*: when correlated features co-occur, a narrow SAE merges them into distributed, partly-polysemantic latents. Hedging is directly relevant to us: it is the most natural mechanism by which a broadly co-occurring concept such as capitalization would be *represented but distributed*, so that a "single dominant latent" is a weak, partly-geometric signal rather than a token-aligned absorber. Notably, Chanin et al. [5] already probe part-of-speech with k-sparse probing, which is why we restrict our novelty claim to the *contrast of the two absorption metrics* on non-spelling properties, not to touching non-spelling properties at all.

SAE evaluation and its reliability. SAEBench [14] standardizes a battery of SAE metrics, including the projection-based absorption score we scrutinize, and documents the layer-decay of the ablation metric that we control for. A growing line questions whether SAE metrics and features are as reliable or as principled as hoped. Closest to our concern, Chanin [3] audits SAEBench's own quality metrics and argues that several should not be trusted as stated—our result is a small, specific instance of that broader worry, aimed at the absorption metric. Leask et al. [16] show SAE features are neither complete nor atomic (the same dictionary re-seeded yields different "units"), and Engels et al. [9] characterize a systematically-missed error component; both undercut the premise, which absorption is defined against, that SAE latents are the canonical units of analysis. On the utility side, Kantamneni et al. [13] find SAE latents do not beat raw-activation linear probes on sparse-probing tasks, situating the downstream "so what" we do not provide inside a live debate about SAE usefulness. Finally, several architectures target absorption directly: Matryoshka SAEs [2] impose a nested feature hierarchy, OrtSAE [15] penalizes decoder cosine similarity, and the recent C2R [11] adds a cross-sample consistency regularizer; whether such mitigations change the projection–ablation disagreement is a natural extension we leave to future work (§8).

Representational vs. causal evidence. Our headline is an instance of a broader methodological point, increasingly emphasized in the field: representational evidence (a probe or projection finding a concept in a direction) need not imply a causal role, and interpretability claims that rest on it can fail to generalize unless grounded causally [12]. Lin and Liu [18] argue, in nearly our terms, that many mechanistic claims report faithfulness/ablation/monosemanticity as *validation* without stating the assumptions that would make them *identifying*—an absorption metric is exactly such a validation, not a causal guarantee. This is the standard motivation for activation patching and causal-mediation analysis (e.g. 19), and it is why the original absorption metric was causal in the first place. Our contribution is the concrete observation that, for a distributed structural feature, the representational absorption signal survives while the causal one vanishes even where the causal metric is valid.

4 Methods

Models, SAEs, and layers. We use Gemma-2-2b at residual-stream layer 12 and Gemma-2-9b at layer 20, each with the Gemma Scope JumpReLU residual SAE of width 16k at an average L_0 near 70–82. Both layers sit well below the depth at which the ablation metric is known to fail. All activations are read at the word-token position of an in-context prompt.

Properties. Concepts are elicited by in-context tasks with objective, tokenizer-derivable labels over the set of single-token, purely-alphabetic vocabulary items ($\approx 1.7 \times 10^5$ tokens). We study two spelling concepts—first-letter and last-letter (26-way)—and four binary structural concepts: is-capitalized (begins with an uppercase letter), all-caps (all uppercase), suffix-ing, and suffix-ed. Only is-capitalized yields a well-powered candidate pool (n=130–273); all-caps and the suffix properties yield n=9–33 and are reported as anecdotes, not evidence. is-numeric was dropped because Gemma's vocabulary contains almost no pure-digit single tokens—itself a label-availability limit worth recording.

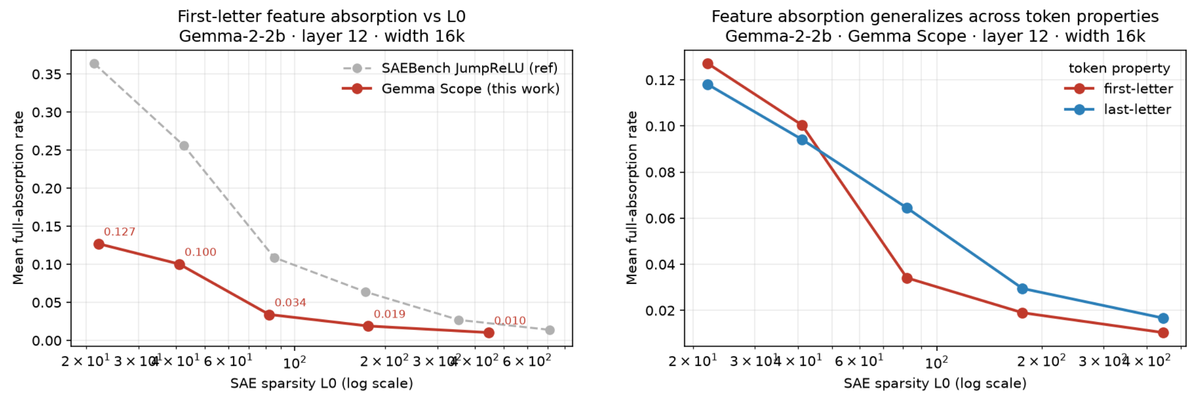
Pipeline. For each concept we (1) train a linear probe on the residual stream at the word position; (2) run k-sparse probing to obtain the main latents M_c via the feature-splitting F_1 criterion; and (3) collect the candidate false negatives—tokens where the probe fires and the true label holds but M_c's sparse probe does not. Absorption is then computed on these candidates by each metric. For binary properties we use a single-class re-implementation of the 26-way pipeline; it was independently reviewed for faithfulness to the reference implementation (a line-by-line check of candidate selection, the denominator, and the absorption predicate).

Ablation metric. We follow Chanin et al. [4]: on each candidate, integrated-gradient attribution of SAE latents to the task logit-difference, counting a token as absorbed if the main latents are silent, the single most-causal latent is clearly dominant (an additive attribution-gap threshold), and it is probe-aligned ($\cos \geq 0.025$). The rate is (absorbed / probe-true-positives). For the binary True/False tasks the metric is the True-minus-False answer-logit.

Projection metric. We compute the per-latent contributions of Eq. (2) at the candidate word position, exclude the main latents, and count a token as single-dominant-absorbed iff $c_{(1)} > 0$ and $c_{(1)} \geq R\, c_{(2)}$. We report the *conditioned* rate over the main-silent pool and a random-direction null (mean over five random unit directions), at $R \in \{1.5, 2, 3\}$.

Task accuracy and controls. For each (property, model) we measure the model's in-context task accuracy. To decide whether an ablation zero means "not causally absorbed" versus one of two confounds, we use: (a) a **cross-model** control —Gemma-2-9b performs is-capitalized far better than 2b; (b) a **layer-matched** spelling control— first-letter ablation absorption at the same 9b layer/L_0; and (c) a **matched-accuracy** control— we sweep first-letter task accuracy by varying the number of in-context examples and, separately, by corrupting a deterministic fraction of in-context answers, so spelling and structural absorption can be compared at equal accuracy.

Statistics and implementation. All rates carry bootstrap 95% confidence intervals (binomial resampling over the candidate pool), and we report candidate counts everywhere because effects are near-floor. Experiments ran on a single NVIDIA A100 (40 GB) via a GPU cluster; Gemma-2-9b runs in bf16. Appendix A gives the full configuration.



(a) Absorption vs. L_0 (Gemma Scope vs. SAEBench reference).

(b) First- vs. last-letter absorption vs. L_0.

Figure 1: Reproduction and within-spelling generalization. (a) The first-letter absorption–L_0 law is recovered on current Gemma Scope SAEs, tracking (below) the SAEBench reference. (b) Last-letter tracks first-letter.

Table 1: Mean absorption rate vs. SAE sparsity L_0 (Gemma-2-2b, layer 12, width 16k). Both spelling properties decrease monotonically with L_0.

L_0	22	41	82	176	445
first-letter	0.127	0.100	0.034	0.019	0.010
last-letter	0.118	0.094	0.065	0.030	0.017

5 Results

5.1 Reproduction and the L0 law

As pipeline validation we first reproduce first-letter absorption on current Gemma Scope SAEs and recover its known dependence on SAE sparsity: absorption rises steeply as L_0 falls (Figure 1a, Table 1), and sits a consistent factor below SAEBench's own JumpReLU numbers—as expected for the more thoroughly trained Gemma Scope SAEs. Last-letter behaves like first-letter (Figure 1b), so within the spelling family the phenomenon is not specific to the first character.

5.2 The projection metric on structural properties

Table 2 and Figure 2 report the projection metric with its random-direction null across dominance thresholds. For spelling, projection absorption is large and far above the null at every R (e.g. 0.53 vs. 0.04 at R=3). For is-capitalized, projection absorption is above the null but *marginal* at n=130 (0.12 at R=3; bootstrap CI on the strict rate $[0.085, 0.200]$ vs. null $[0.008, 0.077]$, Table 7) and roughly 4$\times$ weaker than spelling. The remaining structural properties point the same way but are underpowered (n=9–16) and we do not lean on them. Read alone, the projection metric therefore says: structural concepts show weak-but-present single-latent absorption.

5.3 The ablation metric reports a clean zero

The ablation metric tells the opposite story. For is-capitalized it reports absorption of exactly 0.0—on Gemma-2-2b (with 727 candidate false negatives) and on Gemma-2-9b (0/273 candidates; binomial 95% CI $[0, 0.011]$). The probe and main latent are excellent in both cases (probe AUROC $\approx$ 1.00, k-sparse $F_1 \approx 0.99$), so this is not a failure to find the concept—the concept is

Table 2: Projection (representational) single-dominant-latent rate, conditioned on the main-silent candidate pool, at dominance ratios $R \in \{1.5, 2, 3\}$, with the random-direction null. $*$ marks underpowered pools.

	concept direction			random-direction null			
property	R=1.5	R=2	R=3	R=1.5	R=2	R=3	n
first-letter	0.795	0.670	0.533	0.397	0.174	0.040	2085
last-letter	0.717	0.587	0.454	0.364	0.150	0.033	2308
is-capitalized	0.700	0.423	0.123	0.483	0.209	0.034	130
all-caps*	0.889	0.889	0.333	0.511	0.156	0.000	9
suffix-ing*	0.875	0.875	0.812	0.287	0.175	0.100	16
suffix-ed*	0.750	0.688	0.250	0.350	0.200	0.037	16

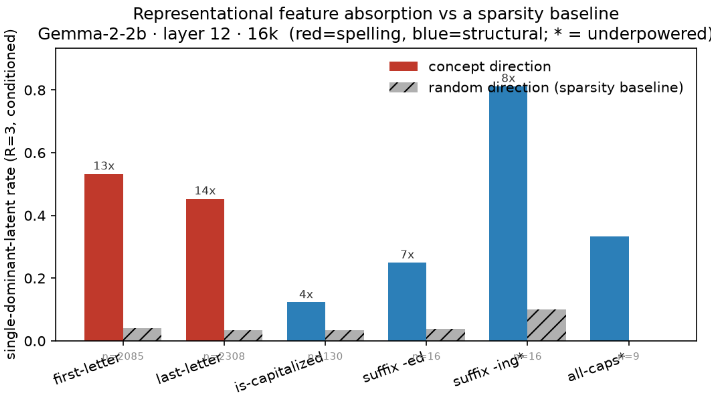


Figure 2: Projection absorption vs. a random-direction sparsity baseline at the strict R=3. Spelling (red) is 13–14× above the null; is-capitalized is ∼3.6× above (robust anchor); the other structural properties are above the null but underpowered ($*$).

cleanly present and cleanly detectable; it simply is not carried causally by a single absorbing latent.

Table 3 shows *why* the ablation metric returns zero, by tracing where candidates drop out of the absorption predicate on 2b. Essentially all sampled candidates have the main latent silent *and* a most-negative-attribution latent (so there is causal structure to test), but almost none have that latent aligned with the concept direction, and—even among those—the attribution is spread across many latents rather than concentrated in one. In other words, when the general capitalization latent misses a token, the model recovers the answer from a diffuse set of small contributions, not from a single token-aligned absorber. This diffuse recovery is exactly what the projection metric's "single dominant latent" criterion counts as (weak) absorption but the causal criterion does not.

5.4 Controls: not task-validity, not a dead layer

The decisive worry is that the ablation zero is the known later-layer artifact rather than a genuine absence of causal absorption. Two controls rule this out (Table 4, Figure 3).

Task ability. Gemma-2-9b performs is-capitalized at 0.91 in-context accuracy, versus 0.60 for Gemma-2-2b (Appendix B, Table 8). If the zero were because the model could not do the task and so provided no behavioral signal to attribute, it should lift on 9b. It does not: 9b is-capitalized ablation absorption is again 0.0, on all 273 candidates.

Layer validity. At the *same* 9b layer and L_0, first-letter ablation absorption is 0.045— higher than 2b's 0.032—so the ablation metric demonstrably fires at that operating point. The zero is

Table 3: Ablation-metric drop-out funnel on Gemma-2-2b (layer 12). For each binary property we show the probe/latent quality, the candidate pool, and how many sampled main-silent candidates pass each successive criterion (most-negative-attribution latent; probe-aligned; both). Almost all candidates have a causal latent, but few are concept-aligned, and the final single-dominant absorption rate is 0.0 throughout.

property	probe AUROC	k-sparse F_1	cand.	main-silent	top-attr. neg.	probe-aligned
is-capitalized	1.000	0.994	727	198/200	198	26
all-caps*	1.000	0.992	23	12/12	12	4
suffix-ing*	1.000	0.867	21	18/18	18	2
suffix-ed*	0.999	0.793	33	24/24	24	2

Table 4: Cross-model ablation (behavioral) absorption. Spelling absorbs on both models; is-capitalized on neither, even as its task accuracy rises from 0.60 (2b) to 0.91 (9b).

property	model / layer	task accuracy	ablation absorption
first-letter	Gemma-2-2b / L12	∼0.95	0.032
first-letter	Gemma-2-9b / L20	∼0.96	0.045
is-capitalized	Gemma-2-2b / L12	0.60	0.000
is-capitalized	Gemma-2-9b / L20	0.91	0.000

therefore specific to the structural property, not the layer.

Matched accuracy. Finally, using the accuracy sweep of §5.5, at matched task accuracy ∼0.9 first-letter ablation absorption is ∼0.026 while is-capitalized is 0.0. So even holding task ability fixed, spelling absorbs and is-capitalized does not.

5.5 The ablation metric is itself task-accuracy-sensitive

The controls above raise a question about the ablation metric in its own right: how much does its reading depend on the model performing the eliciting task? We answer it by manipulating first-letter task accuracy two independent ways and watching both metrics (Tables 5–6, Figure 4). Reducing the number of in-context examples from 10 to 0 drops accuracy from 0.95 to 0.53 and ablation absorption from 0.030 to 0.011; corrupting a growing fraction of in-context answers drops accuracy from 0.97 to 0.80 and ablation absorption from 0.030 to 0.013. In both manipulations ablation absorption tracks accuracy, while the projection rate stays flat at ∼0.60 throughout. The ablation metric thus conflates "the concept is causally absorbed" with "the model performs the task"; the projection metric does not. This both (i) validates our decision to control task accuracy in §5.4 and (ii) is a small, reusable caveat for anyone comparing ablation absorption across settings of differing difficulty.

6 Discussion

The defensible reading is a metric-validity one. For is-capitalized, the SAEBench-standard projection metric's "single dominant latent" criterion reads modestly above chance, yet there is no causal single-latent correlate: the ablation metric—in a regime where it demonstrably works for spelling and where the model performs the task—reports a clean zero. The narrow, checkable claim is therefore that the projection metric appears to *over-report* single-latent absorption for a distributed structural feature whose concept direction is present in the residual but not concentrated in one causally-load-bearing latent. We deliberately avoid the grander phrase "a representational–causal dissociation": the effect is near-floor and rests on one well-powered

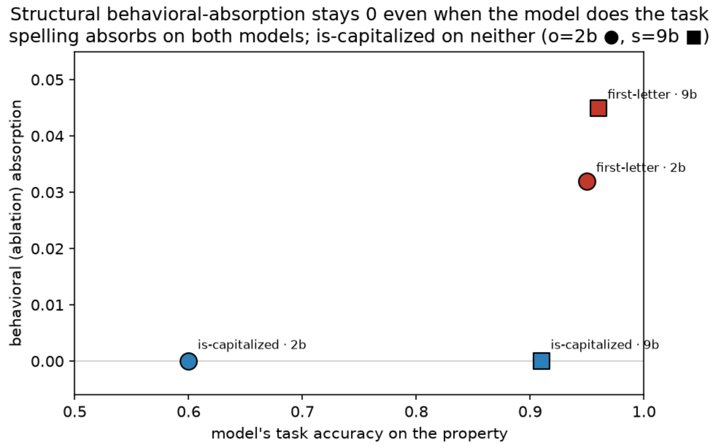


Figure 3: The structural (blue) behavioral-absorption zero persists even as the model's task accuracy rises from 0.60 to 0.91, while spelling (red) absorbs on both models—so the zero is not a task-performance artifact.

Table 5: ICL-count sweep (first-letter, 2b).

#ICL	accuracy	ablation	projection
0	0.53	0.011	0.55
1	0.92	0.035	0.61
3	0.93	0.031	0.60
10	0.95	0.030	0.61

Table 6: Corrupted-ICL sweep (first-letter, 2b).

corrupt	accuracy	ablation	projection
0.00	0.97	0.030	0.62
0.25	0.97	0.026	0.62
0.50	0.93	0.017	0.60
0.75	0.87	0.014	0.60
0.90	0.80	0.013	0.59

property.

The most likely mechanism, stated plainly: feature hedging. Capitalization is a high-frequency, broadly co-occurring concept. Under feature hedging [5], a 16k SAE spreads such a concept across several latents, so that "one dominant latent by projection geometry" is a weak, partly-geometric artifact rather than a token-aligned absorber the model uses. More broadly, some concepts are represented in genuinely multi-latent or multi-dimensional ways [8], in which case a metric premised on a single absorbing latent is simply the wrong instrument, and its "above-null" reading reflects geometry rather than a pathology. Our funnel (Table 3) is consistent with this: when the main latent misses a token, the concept is recovered by many small, mostly-unaligned contributions. We cannot fully separate "the projection metric over-reports" from "hedging makes the projection metric's premise inapplicable here"—but for a practitioner the operational consequence is the same: a projection-flagged single-latent absorption for a distributed feature should not be trusted without causal confirmation.

Implications for SAE evaluation. Absorption scores are used to compare SAEs and to motivate architectures such as Matryoshka [2]. If the projection metric over-reports for distributed features, then absorption comparisons that include non-spelling concepts could reward or penalize SAEs for a partly-geometric signal. Our result is a small piece of evidence for a general prudence already visible in the field [13]: representational SAE metrics benefit from causal cross-checks.

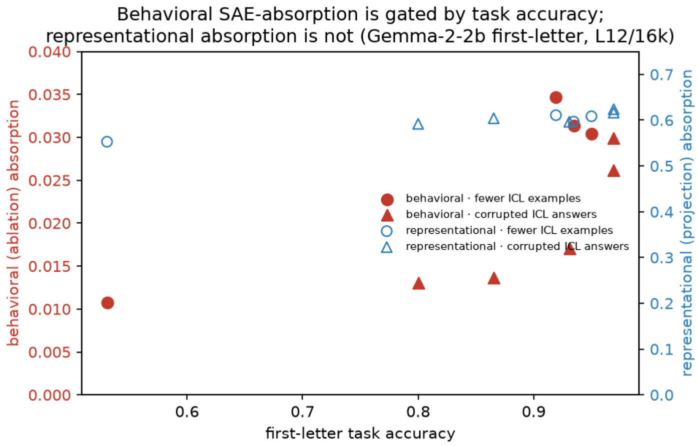


Figure 4: Within first-letter, ablation (red) absorption rises with task accuracy under two independent manipulations (fewer / corrupted in-context examples), while projection (blue) absorption is flat.

7 Limitations and Threats to Validity

We state the weaknesses plainly.

- **Near-floor effects.** Spelling ablation absorption is only ∼0.01–0.05 anywhere, and the is-capitalized projection signal is marginal above its null at n=130. The whole disagreement hinges on distinguishing a few percent from zero; we report CIs, but the numbers are small.
- **One well-powered property.** Only is-capitalized is robust; all-caps and the suffix properties have n=9–33 and are anecdotal. "Property-type-dependent" is effectively an n=1 claim.
- **One model family, one SAE recipe.** Both models are Gemma-2 (cross-*scale* only), all SAEs are Gemma Scope JumpReLU. We run no non-JumpReLU or Matryoshka SAE and no non-Gemma model.
- **No downstream consequence.** We do not exhibit a task in which trusting the projection-flagged absorption would demonstrably mislead—the "so what" that would turn a caveat into an actionable finding.
- **Metric operationalization.** The dominance threshold and the alignment cutoff are choices; we mitigate with an R-sweep and a random null, but a determined skeptic could probe their sensitivity further.

8 Conclusion and Future Work

Measuring SAE feature absorption off the first-letter spelling task exposes a disagreement between its two standard metrics. For is-capitalized, the projection (representational) metric reports weak-but-above-null single-latent absorption that the ablation (causal) metric denies, and controls show the denial is neither a task-performance nor a dead-layer artifact. The most likely mechanism is feature hedging, and the honest headline is a metric-validity caveat: representational absorption signals for distributed structural features can lack a causal correlate.

Three concrete steps would turn this single marginal case into a robust, transferable claim, in rough order of leverage: (1) establish the projection–ablation disagreement on a *second SAE recipe* (e.g. TopK, standard, or Matryoshka) and ideally a second model family, to rule out a Gemma-Scope/JumpReLU artifact and to test whether an absorption-mitigating architecture changes the disagreement; (2) power up a *second* structural property (e.g. is-plural or is-past-tense at $n > 130$), so the effect is property-type-general rather than a capitalization quirk; and (3) exhibit one *downstream decision*—a probing, steering, or unlearning use-case—that the mis-measurement would flip. Until then, we offer the result as a controlled, honestly-scoped caveat rather than a discovery.

Reproducibility. All code, figures, and per-run numbers behind every claim are in the accompanying `absorption-atlas` repository, including the model-agnostic loaders used for the cross-model controls and the independently review-verified single-class absorption pipeline. Appendix A lists the exact configuration.

A Reproducibility details

Models. `google/gemma-2-2b` (26 layers, d=2304) at layer 12; `google/gemma-2-9b` (42 layers, d=3584) at layer 20, loaded via TransformerLens in bf16. **SAEs.** Gemma Scope residual JumpReLU SAEs, width 16k, loaded through SAELens; 2b at $L_0 \approx 82$, 9b at $L_0 \approx 68$ (nearest available to a target of 82). The L0 sweep uses the per-L_0 2b releases at $L_0 \in \{22, 41, 82, 176, 445\}$. **Probes.** One-vs-rest linear probe on `hook_resid_post` activations, 50 epochs, Adam, lr 10^{-2}; binary properties use a single logistic probe. ***k*-sparse probing.** L1 sparse probe over post-ReLU SAE activations; main-latent set grown while validation F_1 jumps by ≥ 0.03. **Ablation metric.** Integrated gradients with 6 interpolation steps; alignment cutoff cos ≥ 0.025; additive attribution-gap threshold 1.0 for the reference 26-way metric (we report a threshold/R sweep for the binary port). **Projection metric.** Contributions of Eq. (2) at the word position; dominance ratios $R \in \{1.5, 2, 3\}$; random-direction null averaged over 5 draws. **Statistics.** Bootstrap 95% CIs, 5000 resamples over the candidate pool. **Hardware.** One NVIDIA A100 40 GB.

B Extended tables

Table 7 gives bootstrap CIs for the strict (R=3) projection rates. Table 8 gives in-context task accuracy for is-capitalized under several prompt formats on both models, showing that 9b performs the task well under every format while 2b is stuck near chance.

Table 7: Strict (R=3) projection absorption with bootstrap 95% CIs vs. the random-direction null.

property	projection [95% CI]	random null [95% CI]	n
first-letter	0.54 [0.51, 0.56]	0.04 [0.03, 0.05]	2085
is-capitalized	0.14 [0.085, 0.200]	0.04 [0.008, 0.077]	130

Table 8: Balanced in-context accuracy on is-capitalized under four prompt formats. Gemma-2-9b performs the task well under every format; Gemma-2-2b is near chance, which is why the 2b ablation zero could a priori be a task-validity artifact—and why the 9b control is needed.

prompt format	Gemma-2-2b	Gemma-2-9b
"… starts with a capital letter: True/False" (random ICL)	0.57	0.89
same, balanced ICL examples	0.59	0.91
"Is … uppercase-initial? Yes/No"	0.63	0.79
"The first letter of … is: uppercase/lowercase"	0.62	0.96

C Hypotheses tested and rejected

Because this project's value is as much in its controls as in its final number, we record the hypotheses we built purpose-made experiments to test and then rejected. Each was discarded on evidence, not taste.

1. **"Absorption is spelling-specific."** An early reading of the raw numbers—structural ablation absorption ≈ 0 vs. spelling ≈ 0.05—suggested a contrarian claim that absorption simply does not generalize beyond character identification. *Rejected:* a task-validity check showed Gemma-2-2b barely performs the binary tasks (is-capitalized accuracy 0.60), so the ablation zero could be a missing-behavioral-signal artifact; and the projection metric with a random-direction null showed structural absorption *is* present above chance. The clean "spelling-specific" claim did not survive.
2. **"The structural zero is a task-validity artifact."** The natural next hypothesis was that ablation reads zero for structural properties only because the model cannot do the task, so a more capable model would show absorption. *Rejected:* Gemma-2-9b performs is-capitalized at 0.91 accuracy yet still shows 0.0 ablation absorption (§5.4); and at matched accuracy spelling still absorbs while is-capitalized does not. Task validity is real *within* spelling (§5.5) but does not explain the structural zero. This falsification is what left the narrower, defensible metric-validity claim.
3. **A denominator/tautology confound.** Because structural main latents are very clean, they miss fewer tokens, giving smaller candidate pools and mechanically lower rates. *Addressed:* we report the rate *conditioned* on the main-silent pool (fraction of missed tokens that absorb), which removes the main-latent-quality confound; the projection-vs-ablation gap survives conditioning.

A methodological reflection: at three points a suggestive positive reading was available and was retired by a control. We consider the controls, not the retired claims, the substantive content.

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